

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 25 — Equilibria

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Acids, Bases & Buffers · pH Calculations · Titration Curves · Solubility Product K_{sp} · Partition Coefficient K_{pc}
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

25.1 Acids, Bases, pH & Buffer Solutions

— Brønsted-Lowry acid-base equilibria; autoionisation of water and ionic product K_w ; K_a , pK_a , K_b , and pK_b

25.2 Acid-Base Titration Curves & Indicators

— Titration curves for strong acid-strong base, weak acid-strong base, strong acid-weak base, and weak acid-weak base

25.3 Solubility Product (K_{sp}) & Common Ion Effect— Solubility equilibria of sparingly soluble ionic compounds; expressions and units of K_{sp} ; calculation of K_{sp} and solubility25.4 Partition Coefficient (K_{pc}) & Solvent Extraction— Distribution of a solute between two immiscible liquid solvents; partition coefficient K_{pc} expression and calculation

High-Frequency Core Repeats (Q41–Q50)

— The 10 most frequently tested Cambridge Paper 4 questions on Equilibria from the past 10 years.

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Weak Acid Titration Curve Analysis & Buffer Region — 9701/42/M/J/23/Q4 [6 Marks]

HARD · Syllabus 25.1

A 25.0 cm^3 sample of $0.100 \text{ mol dm}^{-3}$ ethanoic acid, CH_3COOH ($\text{p}K_{\text{a}} = 4.76$), is titrated against $0.100 \text{ mol dm}^{-3}$ $\text{NaOH}(\text{aq})$. The resulting titration curve is illustrated in Fig. 1.1.

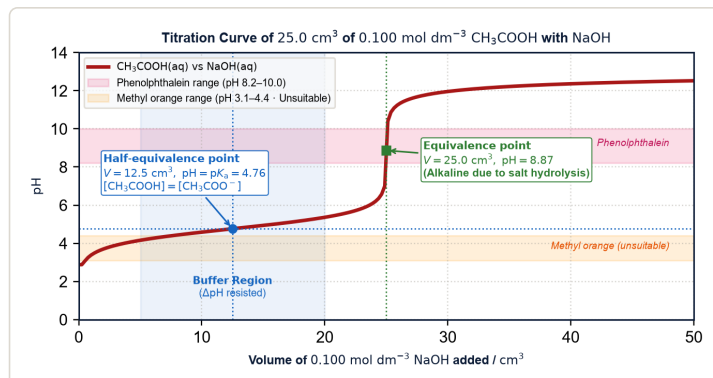


Fig. 1.1: Titration curve of 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ ethanoic acid with $0.100 \text{ mol dm}^{-3}$ NaOH .

- (a) Explain why the pH at the half-equivalence point (12.5 cm^3 NaOH added) equals the $\text{p}K_{\text{a}}$ of ethanoic acid. [2]
- (b) Calculate the pH of the initial $0.100 \text{ mol dm}^{-3}$ CH_3COOH solution prior to adding any NaOH . [2]
- (c) Explain why the pH at the equivalence point is 8.87, writing an ionic equation to account for this alkalinity. [2]

2 Comparison of Four Titration Curves & Indicator Selection — 9701/41/M/J/23/Q4 [6 Marks]

HARD · Syllabus 25.2

The four general types of acid-base titration curves are illustrated in Fig. 2.1.

Available indicators:

- Methyl orange ($\text{pH } 3.1\text{--}4.4$, $\text{p}K_{\text{In}} = 3.7$)
- Bromothymol blue ($\text{pH } 6.0\text{--}7.6$, $\text{p}K_{\text{In}} = 7.0$)
- Phenolphthalein ($\text{pH } 8.3\text{--}10.0$, $\text{p}K_{\text{In}} = 9.3$)

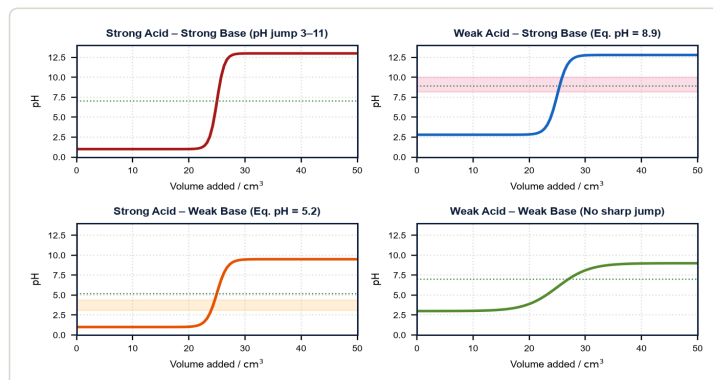


Fig. 2.1: Titration curves for combinations of strong/weak acids and strong/weak bases.

- (a) Identify which indicator(s) from the list can be used for Titration 1 (Strong Acid vs Strong Base), and explain your choice. [2]
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- (b) Identify the most suitable indicator for Titration 3 (Strong Acid vs Weak Base) and explain why phenolphthalein cannot be used. [2]
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- (c) State why no simple acid-base indicator is suitable for Titration 4 (Weak Acid vs Weak Base), and suggest how the end-point can be determined experimentally. [2]
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3 Ethanoic Acid Buffer Solution Preparation & pH Shift — 9701/42/O/N/23/Q4 [6 Marks]

HARD · Syllabus 25.1

A buffer solution is prepared by mixing 400 cm^3 of 0.200 mol dm^{-3} ethanoic acid, CH_3COOH , with 600 cm^3 of 0.150 mol dm^{-3} sodium ethanoate, CH_3COONa .

K_a of $\text{CH}_3\text{COOH} = 1.74 \times 10^{-5}\text{ mol dm}^{-3}$ at 298 K .

- (a) Calculate the pH of this buffer solution. [3]
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- (b) Calculate the change in pH when 5.00 cm^3 of 1.00 mol dm^{-3} HCl is added to 500 cm^3 of this buffer solution. [3]
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4 Blood Hydrogencarbonate Buffer System Mechanism — 9701/41/O/N/23/Q4 [6 Marks]

HARD · Syllabus 25.1

Human arterial blood plasma is maintained within the narrow pH range of 7.35 to 7.45 by the carbonic acid-hydrogencarbonate buffer system illustrated in Fig. 4.1.

Equilibrium: $\text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$, $\text{p}K_a = 6.10$ at 310 K (body temperature).

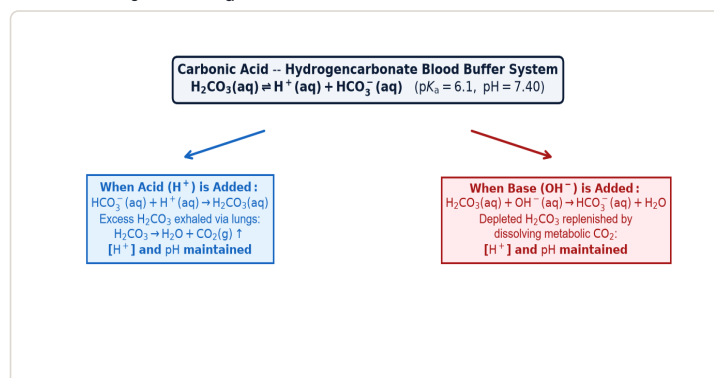


Fig. 4.1: Equilibrium feedback loop of the carbonic acid-hydrogencarbonate blood buffer.

- (a) Calculate the ratio of $[\text{HCO}_3^-]$ to $[\text{H}_2\text{CO}_3]$ required in blood plasma to maintain a normal physiological pH of 7.40. [2]
- (b) With reference to the equilibrium, explain how the blood buffer responds when lactic acid enters the bloodstream during intense muscular exertion. [2]
- (c) Explain why the body's buffer system is an 'open system', and describe the role of the respiratory system in preventing acidosis. [2]

5 Solubility Product and Common Ion Effect on Lead(II) Chloride — 9701/42/M/J/22/Q3 [6 Marks]

HARD · Syllabus 25.3

The solubility product of lead(II) chloride, PbCl_2 , is $1.70 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$ at 298 K.
The equilibrium is shown in Fig. 5.1.

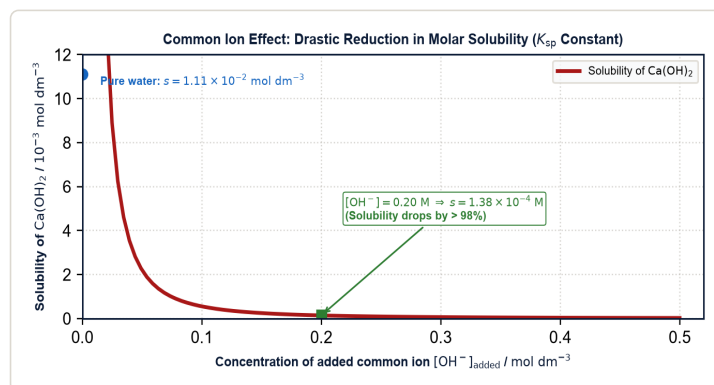


Fig. 5.1: Suppression of PbCl_2 solubility as a function of dissolved chloride concentration.

- (a) Write the expression for the solubility product, K_{sp} , of PbCl_2 , including its units. [2]
- (b) Calculate the molar solubility of PbCl_2 in pure water at 298 K. [2]
- (c) Calculate the molar solubility of PbCl_2 in $0.200 \text{ mol dm}^{-3} \text{ NaCl(aq)}$ at 298 K, and explain the difference using Le Chatelier's principle. [2]

6 Precipitation Feasibility: Ionic Product vs K_{sp} for BaSO_4 — 9701/41/M/J/22/Q3 [6 Marks]

HARD · Syllabus 25.3

A forensic chemist mixes 30.0 cm^3 of $4.00 \times 10^{-4} \text{ mol dm}^{-3} \text{ Ba}(\text{NO}_3)_2$ with 70.0 cm^3 of $2.50 \times 10^{-5} \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4$ at 298 K .
 K_{sp} of $\text{BaSO}_4 = 1.10 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) Calculate the concentration of $\text{Ba}^{2+}(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$ ions in the mixture immediately after mixing, before any possible reaction. [2]

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- (b) Calculate the ionic product of barium sulfate in this mixture. [2]

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- (c) State and explain whether a precipitate of barium sulfate will form under these conditions. [2]

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7 Successive Solvent Extraction of Iodine & Partition Coefficient — 9701/42/O/N/22/Q3 [6 Marks]

HARD · Syllabus 25.4

Iodine distributes between water and cyclohexane according to the partition equilibrium shown in Fig. 7.1:
 $\text{I}_2(\text{aq}) \rightleftharpoons \text{I}_2(\text{cyclohexane})$
At 298 K , the partition coefficient $K_{pc} = [\text{I}_2(\text{cyclohexane})] / [\text{I}_2(\text{aq})] = 85.0$.

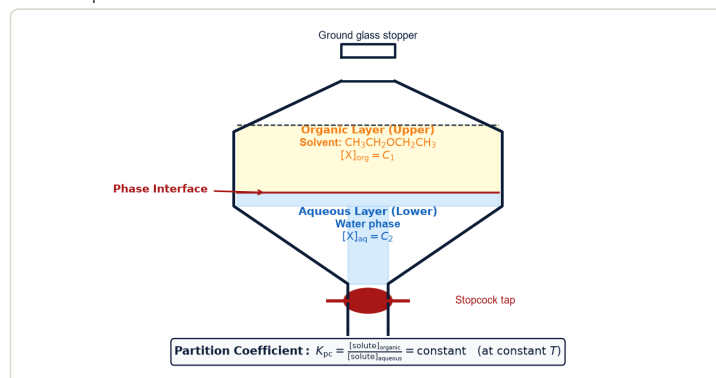


Fig. 7.1: Separating funnel apparatus for the partition of iodine between cyclohexane and water.

- (a) A 100 cm^3 aqueous sample contains 0.0500 g of iodine. Calculate the mass of iodine extracted by shaking with a single 50.0 cm^3 portion of cyclohexane. [3]

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- (b) Alternatively, the same 100 cm^3 sample is extracted using two successive 25.0 cm^3 portions of cyclohexane. Calculate the total mass of iodine extracted and deduce which method is more efficient. [3]

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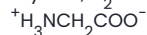
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8 Amino Acid Buffering & Isoelectric Point of Glycine — 9701/41/O/N/22/Q3 [6 Marks]

HARD · Syllabus 25.1

Glycine, $\text{H}_2\text{NCH}_2\text{COOH}$, exists as a zwitterion in aqueous solution:



It has two acid dissociation constants: $\text{pK}_{\text{a1}} = 2.34$ (for $-\text{COOH}$) and $\text{pK}_{\text{a2}} = 9.60$ (for $-\text{NH}_3^+$).

- (a) Draw the structural formula of the zwitterion of glycine and explain why it has a relatively high melting point. [2]

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- (b) Write balanced ionic equations to show how aqueous glycine acts as a buffer against added acid and against added alkali. [2]

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- (c) Calculate the isoelectric point (pI) of glycine, and state the predominant species present at pH 1.0 and at pH 12.0. [2]

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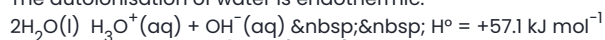
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9 Thermodynamics of K_w and Water Neutrality at Elevated Temperatures — 9701/42/M/J/21/Q2 [6 Marks]

HARD · Syllabus 25.1

The autoionisation of water is endothermic:



At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

At 373 K (100 °C), $K_w = 5.13 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) Explain, in terms of Le Chatelier's principle, why the value of K_w increases as temperature increases. [2]

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- (b) Calculate the pH of pure boiling water at 373 K. [2]

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- (c) A student concludes that boiling water is acidic because its pH is less than 7.00. Evaluate this statement, providing clear chemical reasoning. [2]

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10 Solubility of Calcium Hydroxide and Limewater Titration — 9701/41/M/J/21/Q3 [6 Marks]

HARD · Syllabus 25.3

Limewater is a saturated aqueous solution of calcium hydroxide, $\text{Ca}(\text{OH})_2$.

A 25.0 cm^3 sample of limewater requires 21.6 cm^3 of $0.0500 \text{ mol dm}^{-3}$ HCl for complete neutralisation at 298 K.

- (a) Write the balanced chemical equation for the reaction between $\text{Ca}(\text{OH})_2$ and HCl, and calculate the concentration of $\text{Ca}(\text{OH})_2$ in limewater. [2]

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- (b) Determine the equilibrium concentrations of $\text{Ca}^{2+}(\text{aq})$ and $\text{OH}^{-}(\text{aq})$ in saturated limewater, and calculate K_{sp} for $\text{Ca}(\text{OH})_2$. [2]
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- (c) Calculate the pH of saturated limewater at 298 K ($K_{\text{w}} = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$). [2]
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11 Relative Acid Strengths: Halogenated Carboxylic Acids — 9701/42/O/N/21/Q2 [6 Marks]

HARD · Syllabus 25.1

The table below lists acid dissociation constants at 298 K for four carboxylic acids:

- Ethanoic acid, CH_3COOH : $K_{\text{a}} = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$ ($\text{p}K_{\text{a}} = 4.76$)
- Chloroethanoic acid, CH_2ClCOOH : $K_{\text{a}} = 1.35 \times 10^{-3} \text{ mol dm}^{-3}$ ($\text{p}K_{\text{a}} = 2.87$)
- Dichloroethanoic acid, CHCl_2COOH : $K_{\text{a}} = 5.01 \times 10^{-2} \text{ mol dm}^{-3}$ ($\text{p}K_{\text{a}} = 1.30$)
- Trichloroethanoic acid, CCl_3COOH : $K_{\text{a}} = 2.29 \times 10^{-1} \text{ mol dm}^{-3}$ ($\text{p}K_{\text{a}} = 0.64$)

- (a) Explain the trend in K_{a} across this series in terms of electronic effects and conjugate base stability. [3]
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- (b) Fluoroethanoic acid has $K_{\text{a}} = 2.60 \times 10^{-3} \text{ mol dm}^{-3}$, whereas bromoethanoic acid has $K_{\text{a}} = 1.25 \times 10^{-3} \text{ mol dm}^{-3}$. Explain this difference. [3]
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12 Selective Precipitation of Silver Halides Using K_{sp} — 9701/41/O/N/21/Q3 [6 Marks]

HARD · Syllabus 25.3

A solution contains $0.0100 \text{ mol dm}^{-3} \text{ NaCl}$ and $0.0100 \text{ mol dm}^{-3} \text{ NaI}$.

Aqueous AgNO_3 is slowly added with vigorous stirring.

$K_{\text{sp}}(\text{AgCl}) = 1.80 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$; $K_{\text{sp}}(\text{AgI}) = 8.50 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) Calculate the $[\text{Ag}^{+}]$ required to initiate precipitation of AgI. [2]
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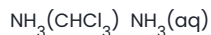
- (b) Calculate the $[\text{Ag}^{+}]$ required to initiate precipitation of AgCl, and deduce which silver halide precipitates first. [2]
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- (c) Calculate the concentration of I^{-} remaining in solution when AgCl just begins to precipitate. [2]
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13 Partition of Ammonia Between Water and Trichloromethane — 9701/42/M/J/20/Q3 [6 Marks]

HARD · Syllabus 25.4

Ammonia distributes between water and trichloromethane, CHCl_3 :



Partition coefficient: $K_{\text{pc}} = [\text{NH}_3(\text{aq})] / [\text{NH}_3(\text{CHCl}_3)] = 25.0$ at 298 K.

Ammonia is titrated against $0.100 \text{ mol dm}^{-3}$ HCl.

- (a) Explain why ammonia is far more soluble in water than in trichloromethane. [2]

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- (b) A 50.0 cm^3 sample of CHCl_3 containing $0.120 \text{ mol dm}^{-3}$ NH_3 is shaken with 100 cm^3 of water until equilibrium is established. Calculate the mass of NH_3 remaining in the organic layer. [2]

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- (c) Calculate the volume of $0.100 \text{ mol dm}^{-3}$ HCl required to neutralise a 25.0 cm^3 aliquot of the aqueous layer. [2]

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14 Buffer Capacity and Dilution Effects — 9701/41/M/J/20/Q4 [6 Marks]

HARD · Syllabus 25.1

Buffer Solution A contains $0.500 \text{ mol dm}^{-3}$ CH_3COOH and $0.500 \text{ mol dm}^{-3}$ CH_3COONa .

Buffer Solution B is prepared by diluting 100 cm^3 of Solution A to 1000 cm^3 with distilled water.

$K_{\text{a}}(\text{CH}_3\text{COOH}) = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$.

- (a) Calculate the pH of Buffer Solution A and Buffer Solution B. [2]

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- (b) Define the term buffer capacity. [2]

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- (c) Compare the buffer capacities of Solutions A and B when 10.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ NaOH is added to 100 cm^3 of each. [2]

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15 pH of Polyprotic Acid Solutions: Carbonic Acid — 9701/42/O/N/19/Q3 [6 Marks]

HARD · Syllabus 25.1

Carbonic acid, H_2CO_3 , is a weak diprotic acid that dissociates in two successive stages at 298 K:

Stage 1: $\text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$ $K_{\text{a1}} = 4.50 \times 10^{-7} \text{ mol dm}^{-3}$

Stage 2: $\text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ $K_{\text{a2}} = 4.80 \times 10^{-11} \text{ mol dm}^{-3}$

- (a) Calculate the pH of a $0.0400 \text{ mol dm}^{-3}$ aqueous solution of carbonic acid. [2]

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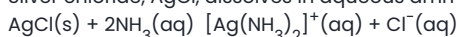
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- (b) State and explain the concentration of $\text{CO}_3^{2-}(\text{aq})$ ions in this solution. [2]
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- (c) Explain why K_{a2} is very much smaller than K_{a1} . [2]
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16 Complex Formation and Silver Halide Dissolution — 9701/41/O/N/19/Q3 [6 Marks]

HARD · Syllabus 25.3

Silver chloride, AgCl, dissolves in aqueous ammonia due to the formation of the diamminesilver(I) complex ion:



Data: $K_{\text{sp}}(\text{AgCl}) = 1.80 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$; Stability constant $K_{\text{stab}}([\text{Ag}(\text{NH}_3)_2]^+) = 1.70 \times 10^7 \text{ dm}^6 \text{ mol}^{-2}$.

- (a) Write the equilibrium expression for the stability constant, K_{stab} , of $[\text{Ag}(\text{NH}_3)_2]^+$. [1]
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- (b) Combine K_{sp} and K_{stab} to show that the overall equilibrium constant K_{c} for the dissolution of AgCl in NH_3 is 3.06×10^{-3} . [2]
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- (c) Explain why AgCl dissolves in dilute $\text{NH}_3(\text{aq})$, whereas AgI does not dissolve even in concentrated $\text{NH}_3(\text{aq})$ ($K_{\text{sp}}(\text{AgI}) = 8.50 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}$). [3]
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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Definition of K_{a} and $\text{p}K_{\text{a}}$ Relationships — 9701/42/M/J/23/Q5 [4 Marks]

MEDIUM · Syllabus 25.1

Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, is a weak monoprotic organic acid with $K_{\text{a}} = 1.38 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K.

- (a) Write the mathematical expression defining $\text{p}K_{\text{a}}$ and calculate $\text{p}K_{\text{a}}$ for lactic acid. [2]
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- (b) State what the relative values of K_{a} and $\text{p}K_{\text{a}}$ indicate about the strength of an acid. [2]
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18 pH of Strong Base Solutions — 9701/41/M/J/23/Q5 [4 Marks]

MEDIUM · Syllabus 25.1

Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong soluble base that fully dissociates in water.
 $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 298 K.

(a) Calculate the concentration of hydroxide ions in $0.0150 \text{ mol dm}^{-3} \text{ Ba}(\text{OH})_2(\text{aq})$. [2]

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(b) Calculate the pH of this solution at 298 K. [2]

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19 Conjugate Acid–Base Pairs in Brønsted–Lowry Theory — 9701/42/O/N/23/Q5 [4 Marks]

MEDIUM · Syllabus 25.1

Consider the reversible equilibrium:
 $\text{HNO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NO}_2^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$

(a) Identify the two conjugate acid–base pairs in this equilibrium. [2]

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(b) Define a Brønsted–Lowry base, and identify which species acts as the base in the reverse direction. [2]

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20 Solubility Product of Aluminium Hydroxide — 9701/41/O/N/23/Q5 [4 Marks]

MEDIUM · Syllabus 25.3

Aluminium hydroxide, $\text{Al}(\text{OH})_3$, is an amphoteric, sparingly soluble solid with $K_{\text{sp}} = 1.00 \times 10^{-33} \text{ mol}^4 \text{ dm}^{-12}$ at 298 K.

(a) Express K_{sp} for $\text{Al}(\text{OH})_3$ in terms of molar solubility, s . [2]

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(b) Calculate the molar solubility s of $\text{Al}(\text{OH})_3$ in pure water at 298 K. [2]

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21 Common Ion Suppression of Magnesium Hydroxide — 9701/42/M/J/22/Q4 [4 Marks]

MEDIUM · Syllabus 25.3

The solubility product of magnesium hydroxide, $\text{Mg}(\text{OH})_2$, is $1.80 \times 10^{-11} \text{ mol}^3 \text{ dm}^{-9}$ at 298 K.

(a) Calculate the solubility of $\text{Mg}(\text{OH})_2$ in pure water in g dm^{-3} ($M_r = 58.3$). [2]

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- (b) Calculate the solubility of $\text{Mg}(\text{OH})_2$ in $0.0500 \text{ mol dm}^{-3} \text{ NaOH(aq)}$. [2]
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22 Partition of Ethanoic Acid Between Benzene and Water — 9701/41/M/J/22/Q4 [4 Marks]

MEDIUM · Syllabus 25.4

Ethanoic acid dissolves in benzene predominantly as dimers due to hydrogen bonding, whereas in water it exists as monomers. Partition data for monomers: $K_{\text{pc}} = [\text{CH}_3\text{COOH(aq)}] / [\text{CH}_3\text{COOH(benzene)}] = 4.20$ at 298 K.

- (a) Draw the structure of the ethanoic acid dimer showing the hydrogen bonds. [2]
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- (b) State two assumptions required for the partition coefficient K_{pc} to remain constant. [2]
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23 Henderson-Hasselbalch Equation for Ammonia Buffer — 9701/42/O/N/22/Q4 [4 Marks]

MEDIUM · Syllabus 25.1

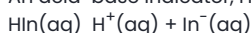
An alkaline buffer contains $0.100 \text{ mol dm}^{-3} \text{ NH}_3$ and $0.200 \text{ mol dm}^{-3} \text{ NH}_4\text{Cl}$.
 $K_{\text{b}}(\text{NH}_3) = 1.80 \times 10^{-5} \text{ mol dm}^{-3}$; $K_{\text{w}} = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) Calculate the concentration of OH^- ions in this buffer solution. [2]
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- (b) Calculate the pH of the buffer solution at 298 K. [2]
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24 Indicator Theory & Transition Range — 9701/41/O/N/22/Q4 [4 Marks]

MEDIUM · Syllabus 25.2

An acid-base indicator, HIn, behaves as a weak acid in aqueous solution:



HIn is yellow and In^- is blue.

$$K_{\text{In}} = 1.00 \times 10^{-5} \text{ mol dm}^{-3}$$

- (a) State the colour of the indicator in strongly acidic solution and in strongly alkaline solution. [2]
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- (b) Explain why the colour changes over the pH range of $\text{p}K_{\text{In}} \pm 1$ (pH 4 to 6). [2]
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25 pH Calculation for Methanoic Acid — 9701/42/M/J/21/Q4 [4 Marks]

MEDIUM · Syllabus 25.1

Methanoic acid, HCOOH , is found in ant venom. At 298 K, $K_a = 1.78 \times 10^{-4} \text{ mol dm}^{-3}$.

- (a) Calculate the pH of $0.0500 \text{ mol dm}^{-3} \text{HCOOH(aq)}$. [2]

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- (b) Calculate the percentage dissociation of methanoic acid in this solution. [2]

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26 Silver Chromate K_{sp} and Solubility in Presence of Chromate Ions — 9701/41/M/J/21/Q4 [4 Marks]

MEDIUM · Syllabus 25.3

The solubility product of silver chromate, Ag_2CrO_4 , is $1.12 \times 10^{-12} \text{ mol}^3 \text{ dm}^{-9}$ at 298 K.

- (a) Calculate the molar solubility of Ag_2CrO_4 in pure water at 298 K. [2]

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- (b) Calculate the molar solubility of Ag_2CrO_4 in $0.100 \text{ mol dm}^{-3} \text{K}_2\text{CrO}_4(\text{aq})$. [2]

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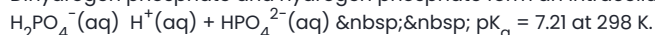
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27 Preparation of Phosphate Buffer Solutions — 9701/42/O/N/21/Q4 [4 Marks]

MEDIUM · Syllabus 25.1

Dihydrogen phosphate and hydrogen phosphate form an intracellular buffer:



- (a) Calculate the ratio of $[\text{HPO}_4^{2-}]$ to $[\text{H}_2\text{PO}_4^-]$ in a solution buffered at pH 7.00. [2]

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- (b) Write an equation showing how this buffer reacts with added H^+ ions. [2]

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28 Partition of Butanedioic Acid Between Ether and Water — 9701/41/O/N/21/Q4 [4 Marks]

MEDIUM · Syllabus 25.4

The partition coefficient of butanedioic acid between diethyl ether and water is:

$$K_{pc} = [\text{acid(ether)}] / [\text{acid(aq)}] = 0.180 \text{ at } 298 \text{ K}$$

- (a) Suggest why butanedioic acid dissolves to a much greater extent in water than in diethyl ether. [2]

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- (b) A solution of 2.00 g of butanedioic acid in 50.0 cm³ of water is shaken with 100 cm³ of ether. Calculate the mass of acid extracted into the ether layer. [2]

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29 Titration of Weak Acid with Strong Base & pH at Half-Neutralisation — 9701/42/M/J/20/Q4 [4 Marks] **MEDIUM · Syllabus 25.2**

During the titration of 20.0 cm³ of 0.100 mol dm⁻³ benzoic acid, C₆H₅COOH (K_a = 6.30 × 10⁻⁵ mol dm⁻³), with 0.100 mol dm⁻³ KOH(aq):

- (a) Calculate the volume of KOH required to reach the half-equivalence point, and state the pH of the mixture at this point. [2]

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- (b) Calculate the pH after the addition of 20.0 cm³ of KOH (at equivalence) given that the concentration of benzoate ions is 0.0500 mol dm⁻³ (K_b = 1.59 × 10⁻¹⁰ mol dm⁻³). [2]

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30 Barium Sulfate Scale Removal by Complexation — 9701/41/M/J/20/Q5 [4 Marks] **MEDIUM · Syllabus 25.3**

Barium sulfate is notorious for forming insoluble scale in offshore oil pipes.

K_{sp}(BaSO₄) = 1.10 × 10⁻¹⁰ mol² dm⁻⁶.

- (a) Explain why flushing the pipes with pure hot water is completely ineffective at removing BaSO₄ scale. [2]

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- (b) Explain how adding aqueous EDTA⁴⁻ (a strong hexadentate chelating ligand) can dissolve the BaSO₄ scale. [2]

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31 Calculation of Kw and Neutral pH at Freezing Point — 9701/42/O/N/19/Q4 [4 Marks] **MEDIUM · Syllabus 25.1**

At 273 K (0 °C), the ionic product of water K_w decreases to 1.14 × 10⁻¹⁵ mol² dm⁻⁶.

- (a) Calculate [H⁺] and the pH of pure water at 273 K. [2]

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- (b) Explain whether water at 273 K is acidic, basic, or neutral. [2]

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32 Partition Coefficient Calculation for Organic Acid — 9701/41/O/N/19/Q4 [4 Marks]

MEDIUM · Syllabus 25.4

An organic acid X has a partition coefficient between ethoxyethane and water of 6.50 at 298 K:

$$K_{pc} = \frac{[X(\text{ethoxyethane})]}{[X(\text{aq})]} = 6.50.$$

- (a) A solution of 0.800 g of X in 80.0 cm³ of water is shaken with 40.0 cm³ of ethoxyethane. Calculate the percentage of X extracted. [2]

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- (b) State what would happen to the apparent value of K_{pc} if the organic acid dimerised in ethoxyethane at higher concentrations. [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Definition of a Brønsted–Lowry Acid & Base — 9701/42/M/J/23/Q6 [2 Marks]

EASY · Syllabus 25.1

The Brønsted–Lowry theory classifies substances by proton transfer.

- (a) Define a Brønsted–Lowry acid. [1]

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- (b) Define a Brønsted–Lowry base. [1]

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34 Definition of Ionic Product of Water K_w — 9701/41/M/J/23/Q6 [2 Marks]

EASY · Syllabus 25.1

Water undergoes self-ionisation in all aqueous systems.

- (a) Write the mathematical expression for K_w . [1]

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- (b) State the units of K_w . [1]

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35 Definition of Buffer Solution — 9701/42/O/N/23/Q6 [2 Marks]

EASY · Syllabus 25.1

Buffer solutions are vital in chemistry and biology.

- (a) Define the term buffer solution. [2]

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36 Definition of Solubility Product K_{sp} — 9701/41/O/N/23/Q6 [2 Marks]

EASY · Syllabus 25.3

Solubility product applies to saturated solutions of sparingly soluble salts.

- (a) Define the term solubility product, K_{sp} . [2]

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37 Common Ion Effect Definition — 9701/42/M/J/22/Q5 [2 Marks]

EASY · Syllabus 25.3

The solubility of salts is modified by the presence of other electrolytes.

- (a) State what is meant by the common ion effect.

[2]

38 Partition Coefficient K_{pc} Definition — 9701/41/M/J/22/Q5 [2 Marks]

EASY · Syllabus 25.4

A solute distributes between two immiscible liquid phases at equilibrium.

- (a) Define the term partition coefficient, K_{pc} .

[2]

39 pH Calculation of a Strong Monoprotic Acid — 9701/42/O/N/22/Q5 [2 Marks]

EASY · Syllabus 25.1

Nitric acid, HNO_3 , is completely dissociated in aqueous solution.

- (a) Calculate the pH of $0.0400 \text{ mol dm}^{-3} \text{HNO}_3(\text{aq})$.

[2]

40 End-Point versus Equivalence Point Distinction — 9701/41/O/N/22/Q5 [2 Marks]

EASY · Syllabus 25.2

Titrations involve both chemical and visual completion points.

- (a) Distinguish between the equivalence point and the end-point of a titration.

[2]

SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Weak Acid–Strong Base Titration & Indicator Range — 9701/42/M/J/23/Q4 [6 Marks]

HARD · Syllabus 25.2

A 25.0 cm^3 sample of $0.100 \text{ mol dm}^{-3}$ propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$ ($K_a = 1.35 \times 10^{-5} \text{ mol dm}^{-3}$), is titrated with $0.100 \text{ mol dm}^{-3} \text{NaOH}$. The titration curve resembles that in Fig. 41.1.

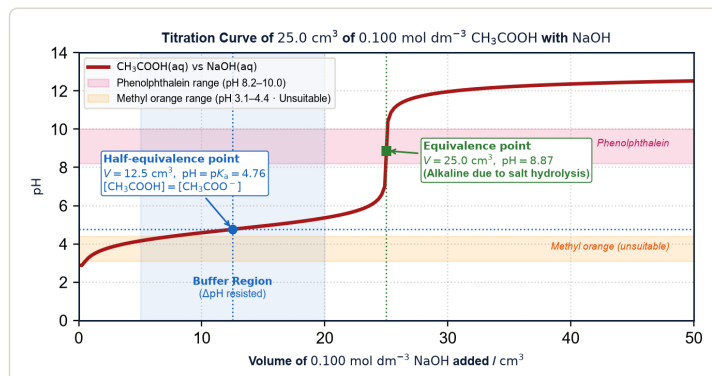


Fig. 41.1: Titration curve of weak acid with strong base showing buffer region and equivalence jump.

- (a) Calculate the initial pH of the propanoic acid solution. [2]
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- (b) State the pH at the half-neutralisation point (12.5 cm³ NaOH added), giving your reasoning. [2]
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- (c) Select a suitable indicator from phenolphthalein (pH 8.3–10.0) and methyl orange (pH 3.1–4.4), explaining why the chosen indicator is effective. [2]
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42 [CORE REPEAT 2] Ethanoate Buffer System Action & pH Calculation — 9701/41/M/J/23/Q4 [6 Marks]

HARD · Syllabus 25.1

A buffer solution is prepared containing 0.250 mol dm⁻³ CH₃COOH and 0.350 mol dm⁻³ CH₃COONa.
K_a(CH₃COOH) = 1.74 × 10⁻⁵ mol dm⁻³.

- (a) Calculate the pH of this buffer solution. [2]
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- (b) Write chemical equations to explain how this buffer maintains a virtually constant pH when small quantities of HCl(aq) or NaOH(aq) are added. [2]
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- (c) Calculate the pH of the solution formed when 10.0 cm³ of 0.200 mol dm⁻³ NaOH is added to 250 cm³ of this buffer. [2]
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43 [CORE REPEAT 3] Solubility Product and Common Ion Precipitation of CaSO₄ — 9701/42/O/N/23/Q4 [6 Marks]

HARD · Syllabus 25.3

The solubility product of calcium sulfate, CaSO₄, is 2.40 × 10⁻⁵ mol² dm⁻⁶ at 298 K.
M_r(CaSO₄) = 136.1.

- (a) Calculate the solubility of CaSO₄ in pure water in g dm⁻³. [2]
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- (b) Calculate the solubility of CaSO₄ in 0.0500 mol dm⁻³ Na₂SO₄(aq). [2]
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- (c) A solution contains $0.0200 \text{ mol dm}^{-3} \text{ Ca}^{2+}$. Calculate the minimum concentration of SO_4^{2-} required to initiate precipitation of CaSO_4 . [2]
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44 [CORE REPEAT 4] Multiple Batch Solvent Extraction Efficiency — 9701/41/O/N/23/Q4 [6 Marks]

HARD · Syllabus 25.4

An organic compound Q has a partition coefficient between ethoxyethane and water of 12.0:

$$K_{pc} = [\text{Q}(\text{ethoxyethane})] / [\text{Q}(\text{aq})] = 12.0.$$

A 50.0 cm^3 aqueous sample contains 0.600 g of Q.

- (a) Calculate the mass of Q extracted by shaking the sample with one 50.0 cm^3 portion of ethoxyethane. [3]
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- (b) Calculate the total mass of Q extracted if the sample is shaken with two successive 25.0 cm^3 portions of ethoxyethane, and explain the chemical principle involved. [3]
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45 [CORE REPEAT 5] Ionic Product vs Ksp Precipitation Calculation — 9701/42/M/J/22/Q3 [4 Marks]

MEDIUM · Syllabus 25.3

Equal volumes of $0.00400 \text{ mol dm}^{-3} \text{ Pb}(\text{NO}_3)_2$ and $0.00200 \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4$ are mixed at 298 K.

$$K_{sp}(\text{PbSO}_4) = 1.80 \times 10^{-8} \text{ mol}^2 \text{ dm}^{-6}.$$

- (a) Calculate the concentration of Pb^{2+} and SO_4^{2-} ions immediately upon mixing. [2]
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- (b) Calculate the ionic product of PbSO_4 and deduce whether a precipitate forms. [2]
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46 [CORE REPEAT 6] pH of Pure Water and Kw at 323 K (50 °C) — 9701/41/M/J/22/Q3 [4 Marks]

MEDIUM · Syllabus 25.1

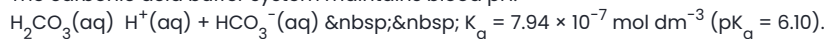
At 323 K (50 °C), the ionic product of water K_w is $5.48 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

- (a) Calculate the pH of pure water at 323 K. [2]
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- (b) Explain whether water at 323 K is acidic, alkaline, or neutral. [2]
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47 [CORE REPEAT 7] Blood Plasma Hydrogencarbonate Buffer Calculations — 9701/42/O/N/22/Q3 [4 Marks] MEDIUM · Syllabus 25.1

The carbonic acid buffer system maintains blood pH:



- (a)** Calculate the $[\text{HCO}_3^-] : [\text{H}_2\text{CO}_3]$ ratio in healthy blood at pH 7.40. **[2]**

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- (b)** State what medical condition arises if the blood pH falls below 7.35, and state which organ regulates $[\text{HCO}_3^-]$. **[2]**

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48 [CORE REPEAT 8] Solubility Product of Silver Halides and Ammonia Test — 9701/41/O/N/22/Q3 [4 Marks] MEDIUM · Syllabus 25.3

Silver chloride (AgCl) and silver bromide (AgBr) have K_{sp} values of 1.80×10^{-10} and $5.00 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$ respectively.

- (a)** Explain why AgCl dissolves in dilute aqueous ammonia while AgBr requires concentrated aqueous ammonia. **[2]**

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- (b)** Write the ionic equation for the complex ion formed when AgCl dissolves in aqueous ammonia. **[2]**

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49 [CORE REPEAT 9] Expression and Units of K_{sp} for Bi_2S_3 — 9701/42/M/J/21/Q3 [2 Marks] EASY · Syllabus 25.3

Bismuth(III) sulfide, Bi_2S_3 , is an extremely insoluble compound.

- (a)** Write the expression for K_{sp} of Bi_2S_3 and deduce its units. **[2]**

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50 [CORE REPEAT 10] Calculation of pH of a Strong Base Solution — 9701/41/M/J/21/Q3 [2 Marks] EASY · Syllabus 25.1

A solution of potassium hydroxide, KOH , has a concentration of $0.0250 \text{ mol dm}^{-3}$ at 298 K.

$$K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}.$$

- (a)** Calculate the pH of this solution. **[2]**

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Mark Scheme — Topic 25 — Equilibria

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	At half-neutralisation, exactly half the CH_3COOH is converted to CH_3COO^- , so $[\text{CH}_3\text{COOH}] = [\text{CH}_3\text{COO}^-]$ [1]; $\ln K_a = [\text{H}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$, the concentrations cancel giving $K_a = [\text{H}^+]$, hence $\text{pH} = \text{p}K_a$ [1].	2
1(b)	$K_a = 10^{-4.76} = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$ [1]; $[\text{H}^+] = \sqrt{K_a \times c} = \sqrt{(1.74 \times 10^{-5} \times 0.100)} = 1.32 \times 10^{-3} \text{ mol dm}^{-3}$ &Arr; $\text{pH} = -\log_{10}(1.32 \times 10^{-3}) = 2.88$ [1].	2
1(c)	All acid is converted to CH_3COO^- ions which hydrolyse water [1]; $\text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$, producing excess OH^- ions [1].	2
2(a)	Both methyl orange and phenolphthalein (or bromothymol blue) are suitable [1]; The vertical pH jump is very long (approx. pH 3.5 to 10.5) and encompasses the working ranges of all three indicators [1].	2
2(b)	Methyl orange is suitable because the vertical jump occurs in the acidic region (pH 3.5 to 7.0) [1]; Phenolphthalein changes colour in the alkaline region (pH 8.3–10.0) where the pH change is gradual before equivalence [1].	2
2(c)	There is no rapid / steep vertical jump in pH at equivalence [1]; Use a pH meter / conductometric titration / temperature titration [1].	2
3(a)	Moles of $\text{CH}_3\text{COOH} = 0.400 \times 0.200 = 0.0800 \text{ mol}$; Moles of $\text{CH}_3\text{COO}^- = 0.600 \times 0.150 = 0.0900 \text{ mol}$ [1]; $[\text{H}^+] = K_a \times (\text{moles acid} / \text{moles base}) = 1.74 \times 10^{-5} \times (0.0800 / 0.0900) = 1.547 \times 10^{-5} \text{ mol dm}^{-3}$ [1]; $\text{pH} = -\log_{10}(1.547 \times 10^{-5}) = 4.81$ [1].	3
3(b)	Moles of H^+ added = $0.0050 \times 1.00 = 0.0050 \text{ mol}$ [1]; In 500 cm ³ buffer: initial moles acid = 0.0400 mol, base = 0.0450 mol; New moles: acid = $0.0400 + 0.0050 = 0.0450 \text{ mol}$, base = $0.0450 - 0.0050 = 0.0400 \text{ mol}$ [1]; New $[\text{H}^+] = 1.74 \times 10^{-5} \times (0.0450 / 0.0400) = 1.958 \times 10^{-5}$ &Arr; new pH = 4.71; $\text{pH} = 4.71 - 4.81 = -0.10$ (falls by 0.10) [1].	3
4(a)	$\text{pH} = \text{p}K_a + \log([\text{HCO}_3^-] / [\text{H}_2\text{CO}_3])$ &Arr; $7.40 = 6.10 + \log(\text{ratio})$ &Arr; $\log(\text{ratio}) = 1.30$ [1]; $[\text{HCO}_3^-] / [\text{H}_2\text{CO}_3] = 10^{1.30} = 20.0$ (or 20 : 1) [1].	2
4(b)	Lactic acid releases H^+ ions which react with the large reservoir of HCO_3^- : $\text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq})$ [1]; Equilibrium shifts to the left, consuming H^+ and preventing a significant drop in pH [1].	2
4(c)	H_2CO_3 is converted into dissolved CO_2 and H_2O : $\text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [1]; Increased ventilation rate (hyperventilation) expels excess CO_2 from the lungs, shifting the equilibria to remove H_2CO_3 and restore normal pH [1].	2
5(a)	$K_{sp} = [\text{Pb}^{2+}][\text{Cl}^-]^2$ [1]; Units: $(\text{mol dm}^{-3})(\text{mol dm}^{-3})^2 = \text{mol}^3 \text{ dm}^{-9}$ [1].	2
5(b)	Let solubility = s; $[\text{Pb}^{2+}] = s$, $[\text{Cl}^-] = 2s$ &Arr; $K_{sp} = 4s^3$ [1]; $s = (1.70 \times 10^{-5} / 4)^{1/3} = (4.25 \times 10^{-6})^{1/3} = 0.0162 \text{ mol dm}^{-3}$ [1].	2
5(c)	$[\text{Cl}^-] \approx 0.200 \text{ mol dm}^{-3}$ &Arr; $K_{sp} = [\text{Pb}^{2+}](0.200)^2 = 0.0400 [\text{Pb}^{2+}]$ &Arr; $[\text{Pb}^{2+}] = 1.70 \times 10^{-5} / 0.0400 = 4.25 \times 10^{-4} \text{ mol dm}^{-3}$ [1]; High $[\text{Cl}^-]$ from NaCl shifts $\text{PbCl}_2(\text{s}) \rightleftharpoons \text{Pb}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$ to the left (common ion effect), drastically reducing solubility [1].	2
6(a)	Total volume = $30.0 + 70.0 = 100.0 \text{ cm}^3$; $[\text{Ba}^{2+}] = 4.00 \times 10^{-4} \times (30.0 / 100.0) = 1.20 \times 10^{-4} \text{ mol dm}^{-3}$ [1]; $[\text{SO}_4^{2-}] = 2.50 \times 10^{-5} \times (70.0 / 100.0) = 1.75 \times 10^{-5} \text{ mol dm}^{-3}$ [1].	2
6(b)	Ionic Product $Q = [\text{Ba}^{2+}][\text{SO}_4^{2-}] = (1.20 \times 10^{-4}) \times (1.75 \times 10^{-5})$ [1]; $Q = 2.10 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$ [1].	2
6(c)	A precipitate of BaSO_4 will form [1]; Because the ionic product Q ($2.10 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$) is greater than K_{sp} ($1.10 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$) [1].	2
7(a)	Let x = mass of I ₂ extracted into 50 cm ³ cyclohexane; $(0.0500 - x)$ remains in 100 cm ³ water; $K_{pc} = (x / 50.0) / ((0.0500 - x) / 100.0) = 85.0$ [1]; $2x / (0.0500 - x) = 85.0$ &Arr; $2x = 4.25 - 85x$ &Arr; $87x = 4.25$ [1]; $x = 0.04885 \text{ g}$ extracted (97.7%) [1].	3
7(b)	First extraction: $(x_1 / 25) / ((0.0500 - x_1) / 100) = 85$ &Arr; $4x_1 = 85(0.0500 - x_1)$ &Arr; $89x_1 = 4.25$ &Arr; $x_1 = 0.04775 \text{ g}$; Remaining = 0.00225 g [1]; Second extraction: $(x_2 / 25) / ((0.00225 - x_2) / 100) = 85$ &Arr; $89x_2 = 85(0.00225) = 0.19125$ &Arr; $x_2 = 0.00215 \text{ g}$ [1]; Total extracted = $0.04775 + 0.00215 = 0.04990 \text{ g}$ (99.8%); Successive extraction with multiple smaller volumes is significantly more efficient than a single batch extraction [1].	3
8(a)	$^+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^-$ drawn correctly [1]; Strong electrostatic ionic attractions between oppositely charged ends in its crystalline lattice require high energy to overcome [1].	2
8(b)	Added acid: $^+\text{H}_3\text{NCH}_2\text{COO}^- + \text{H}^+ \rightleftharpoons ^+\text{H}_3\text{NCH}_2\text{COOH}$ [1]; Added alkali: $^+\text{H}_3\text{NCH}_2\text{COO}^- + \text{OH}^- \rightleftharpoons \text{H}_2\text{NCH}_2\text{COO}^- + \text{H}_2\text{O}$ [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(c)	$pI = (pK_{a1} + pK_{a2}) / 2 = (2.34 + 9.60) / 2 = 5.97$ [1]; At pH 1.0: H_3NCH_2COOH (cation); At pH 12.0: $H_2NCH_2COO^-$ (anion) [1].	2
9(a)	The forward autoionisation reaction is endothermic ($H > 0$) [1]; According to Le Chatelier's principle, increasing temperature shifts the equilibrium in the endothermic direction to absorb added heat, increasing $[H^+]$ and $[OH^-]$ and thus increasing K_w [1].	2
9(b)	$[H^+] = \sqrt{K_w} = \sqrt{(5.13 \times 10^{-13})} = 7.16 \times 10^{-7} \text{ mol dm}^{-3}$ [1]; $pH = -\log_{10}(7.16 \times 10^{-7}) = 6.14$ [1].	2
9(c)	The student's statement is incorrect [1]; The definition of neutrality is $[H^+] = [OH^-]$; in pure water, 1 mol of H^+ is produced for every 1 mol of OH^- , so it remains strictly neutral despite pH 6.14 [1].	2
10(a)	$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$ [1]; Moles $HCl = 0.0216 \times 0.0500 = 1.08 \times 10^{-3} \text{ mol}$ & Arr; Moles $Ca(OH)_2 = 5.40 \times 10^{-4} \text{ mol}$; $[Ca(OH)_2] = 5.40 \times 10^{-4} / 0.0250 = 0.0216 \text{ mol dm}^{-3}$ [1].	2
10(b)	$[Ca^{2+}] = 0.0216 \text{ mol dm}^{-3}$; $[OH^-] = 2 \times 0.0216 = 0.0432 \text{ mol dm}^{-3}$ [1]; $K_{sp} = [Ca^{2+}][OH^-]^2 = (0.0216) \times (0.0432)^2 = 4.03 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$ [1].	2
10(c)	$pOH = -\log_{10}(0.0432) = 1.36$ [1]; $pH = 14.00 - 1.36 = 12.64$ [1].	2
11(a)	Chlorine is electronegative and exerts an electron-withdrawing inductive ($-I$) effect [1]; As the number of chlorine atoms increases from 1 to 3, electron density is pulled away from the carboxylate $-COO^-$ group [1]; This delocalises and disperses the negative charge across the carboxylate anion, stabilising the conjugate base and shifting dissociation to the right [1].	3
11(b)	Fluorine is more electronegative than bromine (4.0 vs 2.8) [1]; Fluorine exerts a significantly stronger electron-withdrawing inductive effect than bromine [1]; The fluoroethanoate anion is more effectively stabilised than the bromoethanoate anion, making fluoroethanoic acid stronger [1].	3
12(a)	$[Ag^+] = K_{sp}(AgI) / [I^-] = 8.50 \times 10^{-17} / 0.0100 = 8.50 \times 10^{-15} \text{ mol dm}^{-3}$ [2].	2
12(b)	$[Ag^+] = K_{sp}(AgCl) / [Cl^-] = 1.80 \times 10^{-10} / 0.0100 = 1.80 \times 10^{-8} \text{ mol dm}^{-3}$ [1]; AgI precipitates first because it requires a much lower concentration of Ag^+ ions [1].	2
12(c)	When $AgCl$ begins to precipitate, $[Ag^+] = 1.80 \times 10^{-8} \text{ mol dm}^{-3}$ [1]; $[I^-] = K_{sp}(AgI) / [Ag^+] = 8.50 \times 10^{-17} / (1.80 \times 10^{-8}) = 4.72 \times 10^{-9} \text{ mol dm}^{-3}$ (over 99.999% of I^- removed) [1].	2
13(a)	Ammonia forms strong hydrogen bonds with water molecules [1]; With $CHCl_3$, ammonia can only form weaker permanent dipole-dipole attractions and London dispersion forces [1].	2
13(b)	Total initial moles $NH_3 = 0.0500 \times 0.120 = 0.00600 \text{ mol}$; Let $y = \text{moles in } CHCl_3$, $(0.00600 - y)$ in water; $K_{pc} = ((0.00600 - y) / 0.100) / (y / 0.0500) = 25.0$ & Arr; $(0.00600 - y) / 2y = 25.0$ & Arr; $0.00600 - y = 50y$ & Arr; $51y = 0.00600$ & Arr; $y = 1.176 \times 10^{-4} \text{ mol}$ [1]; Mass remaining = $1.176 \times 10^{-4} \times 17.0 = 2.00 \times 10^{-3} \text{ g}$ (2.00 mg) [1].	2
13(c)	Moles NH_3 in aqueous layer = $0.00600 - 1.176 \times 10^{-4} = 5.882 \times 10^{-4} \text{ mol}$ in 100 cm^3 & Arr; $[NH_3(aq)] = 0.05882 \text{ mol dm}^{-3}$ [1]; In 25.0 cm^3 aliquot: moles $NH_3 = 0.0250 \times 0.05882 = 1.471 \times 10^{-3} \text{ mol}$ & Arr; Vol $HCl = 1.471 \times 10^{-3} / 0.100 = 0.01471 \text{ dm}^3 = 14.7 \text{ cm}^3$ [1].	2
14(a)	$pH = pK_a + \log([base]/[acid])$; In both solutions, $[base] = [acid]$ so $\log(1) = 0$ & Arr; $pH = pK_a = -\log_{10}(1.74 \times 10^{-5}) = 4.76$ for BOTH Solution A and Solution B [2].	2
14(b)	The amount of acid or base a buffer solution can neutralise before undergoing a significant change in pH [2].	2
14(c)	Solution A has a 10-fold higher buffer capacity than Solution B [1]; Solution A contains 10 times more moles of reserve CH_3COOH and CH_3COO^- per unit volume, resisting pH shift much more effectively [1].	2
15(a)	$[H^+] \approx \sqrt{K_{a1} \times c} = \sqrt{(4.50 \times 10^{-7} \times 0.0400)} = \sqrt{(1.80 \times 10^{-8})} = 1.342 \times 10^{-4} \text{ mol dm}^{-3}$ [1]; $pH = -\log_{10}(1.342 \times 10^{-4}) = 3.87$ [1].	2
15(b)	$K_{a2} = [H^+][CO_3^{2-}] / [HCO_3^-]$; From Stage 1, $[H^+] \approx [HCO_3^-]$ [1]; Cancelling gives $[CO_3^{2-}] \approx K_{a2} = 4.80 \times 10^{-11} \text{ mol dm}^{-3}$ [1].	2
15(c)	It is much more energetically unfavourable to remove a positively charged proton (H^+) from a negatively charged anion (HCO_3^-) than from a neutral molecule (H_2CO_3) due to electrostatic attraction [2].	2
16(a)	$K_{stab} = [Ag(NH_3)_2^+] / ([Ag^+][NH_3]^2)$ [1].	1
16(b)	$K_c = [Ag(NH_3)_2^+][Cl^-] / [NH_3]^2 = K_{sp} \times K_{stab}$ [1]; $K_c = (1.80 \times 10^{-10}) \times (1.70 \times 10^7) = 3.06 \times 10^{-3}$ [1].	2

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16(c)	For AgCl, K_{sp} is sufficiently large (1.8×10^{-10}) that complex formation shifts $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+ + \text{Cl}^-$ to the right by reducing free $[\text{Ag}^+]$ [1]; For AgI, K_{sp} is extremely tiny (8.5×10^{-17}), meaning $[\text{Ag}^+]$ is far too small [1]; The overall K_c for AgI is $(8.5 \times 10^{-17}) \times (1.7 \times 10^{-7}) = 1.4 \times 10^{-23}$, which is negligible even in concentrated NH_3 [1].	3
17(a)	$\text{p}K_a = -\log_{10}(K_a)$ [1]; $\text{p}K_a = -\log_{10}(1.38 \times 10^{-4}) = 3.86$ [1].	2
17(b)	A larger K_a (or smaller/more negative $\text{p}K_a$) indicates greater acid dissociation / stronger acid [2].	2
18(a)	$\text{Ba(OH)}_2 \rightleftharpoons \text{Ba}^{2+} + 2\text{OH}^-$; $[\text{OH}^-] = 2 \times 0.0150 = 0.0300 \text{ mol dm}^{-3}$ [2].	2
18(b)	$\text{pOH} = -\log_{10}(0.0300) = 1.52$ [1]; $\text{pH} = 14.00 - 1.52 = 12.48$ (or $[\text{H}^+] = 10^{-14} / 0.0300 = 3.33 \times 10^{-13}$ & $\text{pH} = 12.48$) [1].	2
19(a)	Pair 1: HNO_2 (acid) and NO_2^- (conjugate base) [1]; Pair 2: H_3O^+ (acid) and H_2O (conjugate base) [1].	2
19(b)	A proton (H^+) acceptor [1]; NO_2^- acts as the base in the reverse direction [1].	2
20(a)	$[\text{Al}^{3+}] = s$, $[\text{OH}^-] = 3s$ & $K_{sp} = [\text{Al}^{3+}][\text{OH}^-]^3 = s(3s)^3 = 27s^4$ [2].	2
20(b)	$s = (K_{sp} / 27)^{1/4} = (1.00 \times 10^{-33} / 27)^{1/4} = (3.704 \times 10^{-35})^{1/4}$ [1]; $s = 2.47 \times 10^{-9} \text{ mol dm}^{-3}$ [1].	2
21(a)	$K_{sp} = 4s^3$ & $s = (1.80 \times 10^{-11} / 4)^{1/3} = 1.651 \times 10^{-4} \text{ mol dm}^{-3}$ [1]; Solubility in $\text{g dm}^{-3} = 1.651 \times 10^{-4} \times 58.3 = 9.63 \times 10^{-3} \text{ g dm}^{-3}$ [1].	2
21(b)	$[\text{OH}^-] \approx 0.0500 \text{ mol dm}^{-3}$ & $K_{sp} = [\text{Mg}^{2+}](0.0500)^2 = 0.0025 [\text{Mg}^{2+}]$ [1]; $[\text{Mg}^{2+}] = s' = 1.80 \times 10^{-11} / 0.0025 = 7.20 \times 10^{-9} \text{ mol dm}^{-3}$ [1].	2
22(a)	Two ethanoic acid molecules drawn facing each other with two parallel dashed $\text{O-H} \cdots \text{O}=\text{C}$ hydrogen bonds forming an 8-membered cyclic dimer [2].	2
22(b)	Constant temperature [1]; The solute exists in the same molecular/chemical state in both solvents (no dissociation or association) / solutions are dilute [1].	2
23(a)	$K_b = [\text{NH}_4^+][\text{OH}^-] / [\text{NH}_3]$ & $[\text{OH}^-] = K_b \times ([\text{NH}_3] / [\text{NH}_4^+])$ [1]; $[\text{OH}^-] = 1.80 \times 10^{-5} \times (0.100 / 0.200) = 9.00 \times 10^{-6} \text{ mol dm}^{-3}$ [1].	2
23(b)	$\text{pOH} = -\log_{10}(9.00 \times 10^{-6}) = 5.05$ [1]; $\text{pH} = 14.00 - 5.05 = 8.95$ [1].	2
24(a)	Acidic: Yellow (high $[\text{H}^+]$ shifts equilibrium left to form HIn) [1]; Alkaline: Blue (added OH^- removes H^+ , shifting equilibrium right to form In^-) [1].	2
24(b)	The human eye can detect a distinct colour transition when the ratio of $[\text{In}^-]$ to $[\text{HIn}]$ varies between 10:1 and 1:10 [1]; By the Henderson equation, $\text{pH} = \text{p}K_{\text{In}} + \log([\text{In}^-]/[\text{HIn}]) = 5 \pm 1$ (pH 4 to 6) [1].	2
25(a)	$[\text{H}^+] = \sqrt{K_a \times c} = \sqrt{(1.78 \times 10^{-4} \times 0.0500)} = \sqrt{(8.90 \times 10^{-6})} = 2.983 \times 10^{-3} \text{ mol dm}^{-3}$ [1]; $\text{pH} = -\log_{10}(2.983 \times 10^{-3}) = 2.53$ [1].	2
25(b)	% dissociation = $([\text{H}^+] / c_{\text{initial}}) \times 100 = (2.983 \times 10^{-3} / 0.0500) \times 100$ [1]; = 5.97% [1].	2
26(a)	$K_{sp} = [\text{Ag}^+]^2[\text{CrO}_4^{2-}] = (2s)^2(s) = 4s^3$ & $s = (1.12 \times 10^{-12} / 4)^{1/3} = 6.54 \times 10^{-5} \text{ mol dm}^{-3}$ [2].	2
26(b)	$[\text{CrO}_4^{2-}] \approx 0.100 \text{ mol dm}^{-3}$ & $K_{sp} = [\text{Ag}^+]^2(0.100) = 1.12 \times 10^{-12}$ & $[\text{Ag}^+]^2 = 1.12 \times 10^{-11}$ [1]; $[\text{Ag}^+] = 3.35 \times 10^{-6} \text{ mol dm}^{-3}$ & solubility = $[\text{Ag}^+] / 2 = 1.67 \times 10^{-6} \text{ mol dm}^{-3}$ [1].	2
27(a)	$\text{pH} = \text{p}K_a + \log([\text{HPO}_4^{2-}] / [\text{H}_2\text{PO}_4^-])$ & $7.00 = 7.21 + \log(\text{ratio})$ & $\log(\text{ratio}) = -0.21$ [1]; ratio = $10^{-0.21} = 0.617$ (or 0.62 : 1) [1].	2
27(b)	$\text{HPO}_4^{2-}(\text{aq}) + \text{H}^+(\text{aq}) \rightleftharpoons \text{H}_2\text{PO}_4^-(\text{aq})$ [2].	2
28(a)	Butanedioic acid has two polar $-\text{COOH}$ groups capable of forming multiple strong hydrogen bonds with water molecules [2].	2
28(b)	$(m / 100) / ((2.00 - m) / 50.0) = 0.180$ & $m / (2(2.00 - m)) = 0.180$ [1]; $m = 0.360(2.00 - m) = 0.720 - 0.360m$ & $1.36m = 0.720$ & $m = 0.529 \text{ g}$ [1].	2
29(a)	Half-neutralisation volume = 10.0 cm^3 [1]; At half-equivalence, $\text{pH} = \text{p}K_a = -\log_{10}(6.30 \times 10^{-5}) = 4.20$ [1].	2

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29(b)	$[\text{OH}^-] = \sqrt{K_b \times [\text{benzoate}]} = \sqrt{1.59 \times 10^{-10} \times 0.0500} = 2.82 \times 10^{-6} \text{ mol dm}^{-3}$ [1]; $\text{pOH} = 5.55$ & $\text{pH} = 14.00 - 5.55 = 8.45$ [1].	2
30(a)	BaSO ₄ has an extremely tiny K _{sp} , making its solubility negligibly small (approx. 10 ⁻⁵ mol dm ⁻³), and its dissolution enthalpy is very small, so hot water dissolves practically none [2].	2
30(b)	EDTA ⁴⁻ binds strongly to Ba ²⁺ ions to form a very stable complex [Ba(EDTA)] ²⁻ with a large stability constant [1]; This dramatically reduces free [Ba ²⁺] in solution, shifting the solubility equilibrium BaSO ₄ (s) ⇌ Ba ²⁺ (aq) + SO ₄ ²⁻ (aq) far to the right until the solid dissolves [1].	2
31(a)	$[\text{H}^+] = \sqrt{K_w} = \sqrt{1.14 \times 10^{-15}} = 3.376 \times 10^{-8} \text{ mol dm}^{-3}$ [1]; $\text{pH} = -\log_{10}(3.376 \times 10^{-8}) = 7.47$ [1].	2
31(b)	It is neutral [1]; Because $[\text{H}^+] = [\text{OH}^-]$ (both are $3.38 \times 10^{-8} \text{ mol dm}^{-3}$); neutrality requires equal concentrations of H ⁺ and OH ⁻ , not necessarily pH = 7.00 [1].	2
32(a)	$(x / 40.0) / ((0.800 - x) / 80.0) = 6.50$ & $2x / (0.800 - x) = 6.50$ & $2x = 5.20 - 6.5x$ & $8.5x = 5.20$ & $x = 0.6118 \text{ g}$ [1]; % extracted = $(0.6118 / 0.800) \times 100 = 76.5\%$ [1].	2
32(b)	The apparent K _{pc} would increase with increasing concentration because dimerization removes monomer X from the organic equilibrium, pulling more solute into the organic layer [2].	2
33(a)	A proton (H ⁺) donor [1].	1
33(b)	A proton (H ⁺) acceptor [1].	1
34(a)	$K_w = [\text{H}^+][\text{OH}^-]$ (or $[\text{H}_3\text{O}^+][\text{OH}^-]$) [1].	1
34(b)	mol ² dm ⁻⁶ [1].	1
35(a)	A solution that resists changes in pH [1]; when small amounts of acid or alkali are added [1].	2
36(a)	The product of the concentrations of each ion present in a saturated solution of a sparingly soluble salt [1]; with each concentration raised to the power of its stoichiometric coefficient in the balanced equilibrium equation [1].	2
37(a)	The reduction in the solubility of a dissolved ionic compound [1]; caused by the addition of a soluble compound that shares an ion in common with it [1].	2
38(a)	The ratio of the concentrations of a solute in two immiscible solvents at equilibrium [1]; at a specified constant temperature [1].	2
39(a)	$[\text{H}^+] = 0.0400 \text{ mol dm}^{-3}$ [1]; $\text{pH} = -\log_{10}(0.0400) = 1.40$ [1].	2
40(a)	Equivalence point is the point at which stoichiometrically equivalent amounts of acid and base have reacted [1]; End-point is the point at which the indicator permanently changes colour [1].	2
41(a)	$[\text{H}^+] = \sqrt{K_a \times c} = \sqrt{1.35 \times 10^{-5} \times 0.100} = 1.162 \times 10^{-3} \text{ mol dm}^{-3}$ [1]; $\text{pH} = -\log_{10}(1.162 \times 10^{-3}) = 2.93$ [1].	2
41(b)	$\text{pH} = \text{pK}_a = -\log_{10}(1.35 \times 10^{-5}) = 4.87$ [1]; At half-neutralisation, [acid] = [salt], so $[\text{H}^+] = K_a$ [1].	2
41(c)	Phenolphthalein is suitable [1]; The vertical pH jump occurs between pH 7.5 and 10.5, perfectly matching the indicator's range (8.3–10.0) [1].	2
42(a)	$[\text{H}^+] = K_a \times ([\text{acid}] / [\text{base}]) = 1.74 \times 10^{-5} \times (0.250 / 0.350) = 1.243 \times 10^{-5} \text{ mol dm}^{-3}$ [1]; $\text{pH} = 4.91$ [1].	2
42(b)	Added H ⁺ : CH ₃ COO ⁻ + H ⁺ ⇌ CH ₃ COOH [1]; Added OH ⁻ : CH ₃ COOH + OH ⁻ ⇌ CH ₃ COO ⁻ + H ₂ O [1].	2
42(c)	Initial moles: acid = $0.250 \times 0.250 = 0.0625 \text{ mol}$; base = $0.250 \times 0.350 = 0.0875 \text{ mol}$; Moles OH ⁻ added = $0.010 \times 0.200 = 0.0020 \text{ mol}$; New moles: acid = 0.0605 mol , base = 0.0895 mol [1]; New $[\text{H}^+] = 1.74 \times 10^{-5} \times (0.0605 / 0.0895) = 1.176 \times 10^{-5}$ & $\text{pH} = 4.93$ [1].	2
43(a)	$s = \sqrt{K_{sp}} = \sqrt{2.40 \times 10^{-5}} = 4.90 \times 10^{-3} \text{ mol dm}^{-3}$ [1]; Solubility in g dm ⁻³ = $4.90 \times 10^{-3} \times 136.1 = 0.667 \text{ g dm}^{-3}$ [1].	2
43(b)	$[\text{SO}_4^{2-}] \approx 0.0500 \text{ mol dm}^{-3}$ & $[\text{Ca}^{2+}] = s' = K_{sp} / 0.0500 = 2.40 \times 10^{-5} / 0.0500$ [1]; $s' = 4.80 \times 10^{-4} \text{ mol dm}^{-3}$ [1].	2
43(c)	$[\text{SO}_4^{2-}] = K_{sp} / [\text{Ca}^{2+}] = 2.40 \times 10^{-5} / 0.0200$ [1]; $[\text{SO}_4^{2-}] = 1.20 \times 10^{-3} \text{ mol dm}^{-3}$ [1].	2

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44(a)	$(m / 50.0) / ((0.600 - m) / 50.0) = 12.0$ &Arr; $m / (0.600 - m) = 12.0$ [1]; $m = 7.20 - 12m$ &Arr; $13m = 7.20$ &Arr; $m = 0.554$ g (92.3%) [2].	3
44(b)	1st extraction: $(m_1 / 25.0) / ((0.600 - m_1) / 50.0) = 12.0$ &Arr; $2m_1 / (0.600 - m_1) = 12.0$ &Arr; $14m_1 = 7.20$ &Arr; $m_1 = 0.5143$ g; Remaining = 0.0857 g [1]; 2nd extraction: $2m_2 / (0.0857 - m_2) = 12.0$ &Arr; $14m_2 = 1.0286$ &Arr; $m_2 = 0.0735$ g; Total = 0.5143 + 0.0735 = 0.588 g (98.0%) [1]; Multiple extractions using smaller volumes maintain a larger concentration gradient in each cycle, removing a greater total fraction of solute [1].	3
45(a)	Due to equal volume mixing, dilution factor = 2 &Arr; $[Pb^{2+}] = 0.00200$ mol dm ⁻³ ; $[SO_4^{2-}] = 0.00100$ mol dm ⁻³ [2].	2
45(b)	Ionic product $Q = [Pb^{2+}][SO_4^{2-}] = (2.00 \times 10^{-3}) \times (1.00 \times 10^{-3}) = 2.00 \times 10^{-6}$ mol ² dm ⁻⁶ [1]; Since $Q (2.00 \times 10^{-6}) > K_{sp} (1.80 \times 10^{-8})$, a white precipitate of PbSO ₄ forms [1].	2
46(a)	$[H^+] = \sqrt{K_w} = \sqrt{(5.48 \times 10^{-14})} = 2.341 \times 10^{-7}$ mol dm ⁻³ [1]; $pH = -\log_{10}(2.341 \times 10^{-7}) = 6.63$ [1].	2
46(b)	Neutral [1]; Because $[H^+] = [OH^-]$; water is only acidic if $[H^+] > [OH^-]$ [1].	2
47(a)	$pH = pK_a + \log([HCO_3^-]/[H_2CO_3])$ &Arr; $7.40 = 6.10 + \log(\text{ratio})$ &Arr; $\log(\text{ratio}) = 1.30$ [1]; $\text{ratio} = 10^{1.30} = 20 : 1$ [1].	2
47(b)	Acidosis [1]; The kidneys regulate $[HCO_3^-]$ by reabsorption / excretion [1].	2
48(a)	AgBr has a much lower K_{sp} than AgCl, producing a much lower equilibrium $[Ag^+]$ [1]; Formation of $[Ag(NH_3)_2]^+$ requires a higher $[NH_3]$ to lower $[Ag^+]$ sufficiently to shift the dissolution equilibrium of AgBr to the right [1].	2
48(b)	$AgCl(s) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$ [2].	2
49(a)	$K_{sp} = [Bi^{3+}]^2[S^{2-}]^3$ [1]; Units = (mol dm ⁻³) ⁵ = mol ⁵ dm ⁻¹⁵ [1].	2
50(a)	$pOH = -\log_{10}(0.0250) = 1.60$ [1]; $pH = 14.00 - 1.60 = 12.40$ [1].	2